WOLLO UNIVERSITY

KOMBOLCHA INSTITUTE OF TECHNOLOGY

COLLEGE OF INFORMATICS

DEPARTMENT OF INFORMATION TECHNOLOGY

Course Title: Network Design (ITec4111)

Target Group: 4th Year Information Technology Students (Section A & B)

Instructor: Gashaw W.

Assessment Weight: Semester Project — 20%

Semester Project Assignment Guidelines

1. **Overview**

As part of the practical requirements for Network Design (ITec4111), students are required to complete a comprehensive enterprise network design project. This project evaluates your ability to apply core networking concepts—including switching design, IP addressing, routing, WAN topologies, and wireless integration—to solve realistic infrastructure challenges.

Students may work individually or in assigned groups using simulation tools (such as Cisco Packet Tracer, GNS3, or EVE-NG).

2. Approved Project Titles(KIOT Focus)

Choose one of the following general project topics for your implementation and report:

1. **KIOT Campus Network and Switching Design**

   Description: Design a reliable and scalable campus network infrastructure for KIOT featuring secure switching configurations, traffic segmentation (VLANs), and loop-prevention mechanisms (Spanning Tree Protocol).

2. **KIOT IPv4 Addressing and Dynamic Routing Implementation**

   Description: Develop an optimized IP addressing and subnetting scheme combined with dynamic routing protocols to ensure efficient data flow and routing scalability across KIOT's multiple network segments(Academic blocks, offices and laboratories).

3. **KIOT Multi-Site WAN and Remote Connectivity Infrastructure Design**

   Description: Architect a multi-site Wide Area Network that connects KIOT's main campus facilities to external branch connections or administrative annexes using structured topologies (Star, Tree, or Mesh) and reliable remote access links.

4. **KIOT Enterprise Wireless LAN and Security Architecture**

   Description: Plan a secure and efficient wireless network coverage plan across KIOT campus zones, incorporating proper channel allocation, authentication, and wireless management strategies for students and staff

5. **Comprehensive KIOT Enterprise Network and Security Design**

   Description: Develop an end-to-end enterprise network architecture for KIOT that integrates secure switching, backbone connectivity, data center/server farm access, and robust multi-layered security controls

3. Project Deliverables

Your final submission must include:

- Project Proposal: Brief description of the scenario, business requirements, and proposed layout.
- Topology Diagrams: Clear physical and logical network diagrams showing all connected devices, links, and subnet allocations.
- Configuration / Simulation Files: Fully working configuration scripts or simulation project files.
- Final Technical Report: Complete documentation detailing the design process, configuration steps, testing/verification results, and challenges faced.